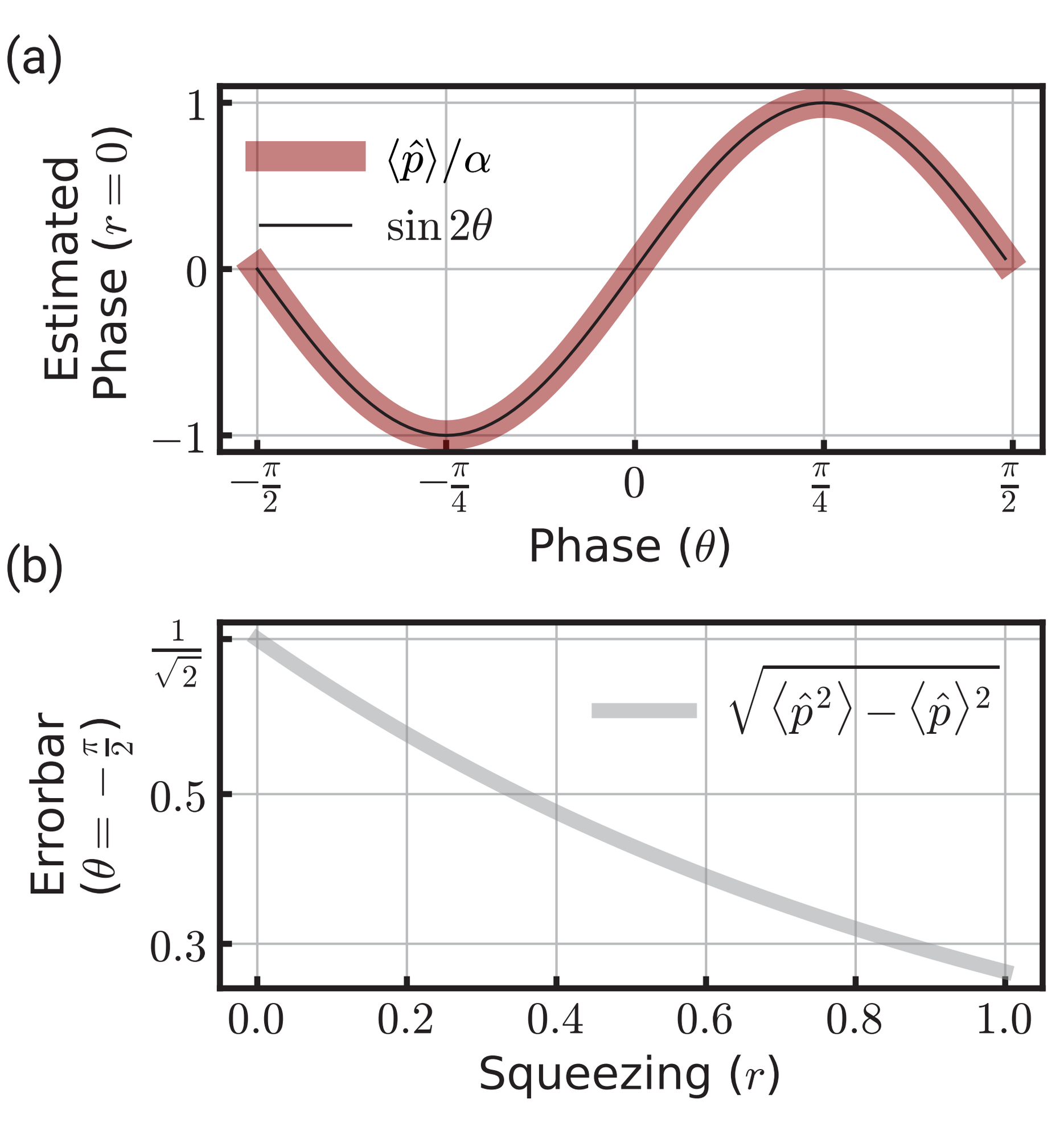

(a)
Estimated Phase ($r = 0$)
1
0
-1
$\langle \hat{p} \rangle / \alpha$
$\sin 2\theta$
$-\frac{\pi}{2}$
$-\frac{\pi}{4}$
0
$\frac{\pi}{4}$
$\frac{\pi}{2}$
Phase (θ)
(b)
Errorbar ($\theta = -\frac{\pi}{2}$)
$\frac{1}{\sqrt{2}}$
0.5
0.3
$\sqrt{\langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2}$
0.0
0.2
0.4
0.6
0.8
1.0
Squeezing (r)